

Rules of Differentiation (58)

1. Derivative of a constant is 0.

proof

Let $f(x) = c$.

$$\Rightarrow f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{c - c}{\Delta x} = 0$$

2. Derivative of a linear function $ax+b$

proof

Let $f(x) = ax+b$.

$$\Rightarrow f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{a(x+\Delta x)+b - ax-b}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{a\Delta x}{\Delta x} = \lim_{\Delta x \rightarrow 0} a = a.$$

3. Derivative of a power function (power Rule).

If n a positive integer, then

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

proof.

Let $f(x) = x^n$.

$$\text{then } f(x+\Delta x) = (x+\Delta x)^n = x^n + nx^{n-1}\Delta x + \dots + nx^{n-1}\Delta x + \Delta x^n$$

$$\Rightarrow f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{nx^{n-1}\Delta x + \dots + nx^{n-1}\Delta x + \Delta x^n}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} nx^{n-1} + \dots + nx^{n-1} = nx^{n-1}.$$

This is because every term after the first contains Δx .

$\Rightarrow f'(x)$ of $f(x) = x^n$ is nx^{n-1} .

4. Constant Multiple Rule.

If $f(x)$ is differentiable and k is a constant, then

$$\frac{d}{dx} kf(x) = k \frac{d}{dx} f(x).$$

$$5. \frac{d}{dx} [f(x) + g(x)] = \frac{d}{dx} f(x) + \frac{d}{dx} g(x).$$

$$\frac{d}{dx} [f(x) - g(x)] = \frac{d}{dx} f(x) - \frac{d}{dx} g(x).$$

Nb: This rule can also be generalised to a finite set of functions

6. The product Rule.

Consider the function $y = f(x) = 3x^2 \sin x$

This function is a product of two functions
i.e. $3x^2$ and $\sin x$

Let $u = 3x^2$ and $v = \sin x \Rightarrow y = u \cdot v$.

Let x increase by a small increment Δx ,
causing incremental change in u , v and
 y of Δu , Δv and Δy respectively.

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Then $y = 3x^2 \cdot \sin x$
 $= u \cdot v$

$$\Rightarrow (y + \Delta y) = (u + \Delta u)(v + \Delta v)$$

$$= uv + u\Delta v + v\Delta u + \Delta u \Delta v$$

$$\Rightarrow (y + \Delta y) - y = u\Delta v + v\Delta u + \Delta u \Delta v$$

$$\Rightarrow \frac{\Delta y}{\Delta x} = \frac{u\Delta v + v\Delta u + \Delta u \Delta v}{\Delta x}$$

As $\Delta x \rightarrow 0$, then $\Delta u \rightarrow 0$, $\Delta v \rightarrow 0$ and $\Delta y \rightarrow 0$

$$\Rightarrow \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} u \cdot \frac{\Delta v}{\Delta x} + \lim_{\Delta x \rightarrow 0} v \cdot \frac{\Delta u}{\Delta x} + \lim_{\Delta x \rightarrow 0} \frac{\Delta u \cdot \Delta v}{\Delta x}$$

but $\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \frac{dy}{dx}$

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta v}{\Delta x} = \frac{dv}{dx}$$

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta u}{\Delta x} = \frac{du}{dx}$$

and $\lim_{\Delta x \rightarrow 0} \frac{\Delta u \cdot \Delta v}{\Delta x} = 0$

$$\Rightarrow \frac{dy}{dx} = u \cdot \frac{dv}{dx} + v \cdot \frac{du}{dx} \quad \Bigg| \text{The product Rule.}$$

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$$\Rightarrow y = u(x) \cdot v(x)$$

$$\frac{dy}{dx} = u \cdot v'(x) + v'(x)u'(x)$$

$$\text{i.e. } f'(x) = u'v + v'u.$$

$$\Rightarrow \text{If } k(x) = f(x) \cdot g(x)$$

$$\text{Then } k'(x) = f'(x)g(x) + g'(x)f(x).$$

7. Quotient Rule.

$$\text{If } k(x) = \frac{f(x)}{g(x)} : g(x) \neq 0, \text{ then}$$

$$k'(x) = \frac{f'(x)g(x) - g'(x)f(x)}{[g(x)]^2}$$

$$\text{i.e. If } f(x) = \frac{u(x)}{v(x)},$$

$$\text{then } f'(x) = \frac{u'(x)v(x) - v'(x)u(x)}{(v(x))^2}$$

$$\text{i.e. } \frac{u'v - v'u}{v^2}.$$

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For example, if $f(x) = \frac{3x^2+2x}{2x^2+1}$.

$$\text{Then } f'(x) = \frac{u'v - v'u}{v^2} = \frac{(6x+2)(2x^2+1) - 4x(3x^2+2x)}{(2x^2+1)^2}.$$

8. Chain Rule.

If $y = f(u)$ and $u = g(x)$, and the derivatives $\frac{dy}{du}$ and $\frac{du}{dx}$ both exist,

then the composite function defined by $y = f(g(x))$ has the derivative

given by

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}.$$

Example:

1. If $y = (3t^4 - 2t)^5$, find $\frac{dy}{dt}$.

Solution

Let $u = 3t^4 - 2t \Rightarrow y = u^5$.

$$\begin{aligned} \Rightarrow \frac{dy}{dt} &= \frac{dy}{du} \cdot \frac{du}{dt} = 5u^4 (12t^3 - 2) \\ &= 5(3t^4 - 2t)^4 (12t^3 - 2). \end{aligned}$$

2. If $y = (4x-3)^7$, find $\frac{dy}{dx}$.

Solution

Let $u = 4x-3 \Rightarrow y = u^7$.

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = 7u^6 \cdot 4 = 28(4x-3)^6.$$

Derivatives of Trigonometric function (63)

$$\Delta x \sin x = \cos x$$

$$\Delta x \cos x = -\sin x$$

proof

Let $f(x) = \sin x$, then

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\sin(x+\Delta x) - \sin x}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\sin x \cos \Delta x + \cos x \sin \Delta x - \sin x}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\sin x \cos \Delta x - \sin x + \cos x \sin \Delta x}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \sin x \left[\frac{\cos \Delta x - 1}{\Delta x} \right] + \lim_{\Delta x \rightarrow 0} \cos x \frac{\sin \Delta x}{\Delta x}$$

$$\text{but } \lim_{\Delta x \rightarrow 0} \frac{\cos \Delta x - 1}{\Delta x} = 0 \text{ and } \lim_{\Delta x \rightarrow 0} \frac{\sin \Delta x}{\Delta x} = 1.$$

$$\Rightarrow f'(x) = \sin x \cdot 0 + \cos x \cdot 1 = \cos x.$$

NB: Using $\cos(x+\Delta x) = \cos x \cos \Delta x - \sin x \sin \Delta x$,

then it can be shown that
 $\Delta x \cos x = -\sin x$.

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$$* \frac{d}{dx} \tan x = \frac{d}{dx} (\tan x) = \frac{d}{dx} \frac{\sin x}{\cos x}$$

To find the derivative of $\tan x = \frac{\sin x}{\cos x}$

We use the quotient Rule.

Let $f(x) = \tan x = \frac{\sin x}{\cos x}$, then

$$f'(x) = \frac{\sin x' \cos x - \cos x' \sin x}{[\cos x]^2}$$

$$= \frac{\cos x \cos x - (-\sin x) \sin x}{\cos^2 x}$$

$$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x}$$

$$\frac{1}{\cos^2 x} = \sec^2 x$$

$$\text{but } \frac{1}{\cos x} = \sec x \Rightarrow \frac{d}{dx} \tan x = \sec^2 x$$

$$* \frac{d}{dx} \sec x = \frac{d}{dx} \frac{1}{\cos x}$$

Let $f(x) = \sec x = \frac{1}{\cos x}$

$$\Rightarrow f'(x) = \frac{1' \cos x - \cos x' \cdot 1}{[\cos x]^2} = \frac{0 - (-\sin x)}{[\cos x]^2}$$

$$= \frac{\sin x}{\cos^2 x} = \frac{1}{\cos x} \times \frac{\sin x}{\cos x}$$

$$= \sec x \tan x$$

Derivatives of Trigonometric Functions (65)

$D_x \sin x$	$\cos x$
$D_x \cos x$	$-\sin x$
$D_x \tan x$	$\sec^2 x$
$D_x \sec x$	$\sec x \tan x$
$D_x \operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$
$D_x \cot x$	$-\operatorname{cosec}^2 x$

NB:

If the coefficient of x is different from one, then the derivative of the function has to cater for that.

⇒ If a is a constant, then

$$D_x \sin(ax) = a \cos(ax)$$

$$D_x \cos(ax) = -a \sin(ax)$$

$$D_x \tan(ax) = a \sec^2(ax)$$

$$D_x \sec(ax) = a \sec(ax) \tan(ax)$$

$$D_x \operatorname{cosec}(ax) = -a \operatorname{cosec}(ax) \cot(ax)$$

$$D_x \cot(ax) = -a \operatorname{cosec}^2(ax)$$

Derivative of an exponential function (26)

If $f(x) = e^x$, then $f'(x) = e^x$.

If $f(x) = e^{ax}$, then $f'(x) = ae^{ax}$ where a is a constant.

For example:

If $f(x) = e^{3x}$, then $f'(x) = 3e^{3x}$.

Note: If $y = f(x) = e^{3x^2}$, then we use the chain Rule

i.e. Let $u = 3x^2 \Rightarrow \frac{du}{dx} = 6x$.

$$y = e^u \Rightarrow \frac{dy}{du} = e^u$$

$$\Rightarrow \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = 6x e^u = 6x e^{3x^2}$$

Derivative of a Logarithmic function

If $f(x) = \ln x$, then $f'(x) = \frac{1}{x}$.

If $f(x) = \ln(ax)$, then $f'(x) = \frac{1}{x}$, where a is a constant.

Note:

If $y = f(x) = \ln(ax) = \ln a + \ln x$.

$$\Rightarrow f'(x) = 0 + \frac{1}{x} = \frac{1}{x}$$

Since $\ln a$ is a constant; the derivative of a constant is zero.

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If $f(x) = \ln(x+1)$, then we use the Substitution Method (also called the Chain Rule).

$$\Rightarrow f(x) = \ln(x+1)$$

$$\text{Let } u = x+1 \Rightarrow \frac{du}{dx} = 1$$

$$y = f(x) = \ln u \Rightarrow \frac{dy}{du} = \frac{1}{u}$$

$$\Rightarrow \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = \frac{1}{u} = \frac{1}{x+1}$$

$\Rightarrow$ If $f(x) = \ln(x^2+x)$, then

$$f'(x) = \frac{2x+1}{x^2+x}$$

Examples:

1. If $y = \sin(x^2+2)$, find y' .

Solution

$$y = \sin(x^2+2)$$

$$\text{Let } u = x^2+2$$

$$y = \sin u$$

$$\Rightarrow \frac{dy}{du} = \cos u \quad \text{and} \quad \frac{du}{dx} = 2x$$

$$\Rightarrow \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = \cos u \cdot 2x = 2x \cos(x^2+2)$$

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2. If $y = \tan x \sec x$, find $\frac{dy}{dx}$.

Solution

$$y = \tan x \times \sec x \Rightarrow \text{product Rule.}$$

$$\text{Let } u = \tan x \Rightarrow u' = \sec^2 x.$$

$$v = \sec x \Rightarrow v' = \sec x \tan x.$$

$$\begin{aligned} \Rightarrow y' &= u'v + v'u \\ &= \sec^2 x \cdot \sec x + \sec x \tan x \cdot \tan x \\ &= \sec^3 x + \sec x \tan^2 x \\ &= \sec x (\sec^2 x + \tan^2 x) \end{aligned}$$

3. If $y = \frac{\sin x}{1 + \cos x}$.

Solution

$$y = \frac{\sin x}{1 + \cos x} = \frac{u}{v} \Rightarrow \text{Quotient Rule.}$$

$$u = \sin x \Rightarrow u' = \cos x$$

$$v = 1 + \cos x \Rightarrow v' = -\sin x$$

$$y' = \frac{u'v - v'u}{v^2} = \frac{\cos x(1 + \cos x) - (-\sin x)\sin x}{[1 + \cos x]^2}$$

$$= \frac{\cos x + \cos^2 x + \sin^2 x}{[1 + \cos x]^2}$$

$$= \frac{\cos x + 1}{[1 + \cos x]^2} = \frac{1}{1 + \cos x}.$$

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4. If $f(x) = (\tan x - x)^2$, find $f'(x)$ Solution

$$f(x) = (\tan x - x)^2$$

$$\text{Let } u = \tan x - x \Rightarrow \frac{du}{dx} = \sec^2 x - 1$$

$$y = u^2 \Rightarrow \frac{dy}{du} = 2u$$

$$\Rightarrow f'(x) = \frac{dy}{dx} = 2u \cdot (\sec^2 x - 1)$$

$$= 2[\tan x - x][\sec^2 x - 1]$$

5. If $y = \tan^2(2x+3)$, show that

$$y' = 2 \tan(2x+3) \cdot 2 \sec^2(2x+3)$$

$$= 4 \tan(2x+3) \sec^2(2x+3)$$

6. If $y = f(x) = \sec\left(\frac{1}{x}\right)$, find y' Solution

$$y = \sec\left(\frac{1}{x}\right)$$

$$\text{Let } u = \frac{1}{x} = x^{-1} \Rightarrow \frac{du}{dx} = -x^{-2} = -\frac{1}{x^2}$$

$$y = \sec u, \text{ then } \frac{dy}{du} = \sec u \tan u$$

$$\Rightarrow \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = -\frac{1}{x^2} \sec u \tan u$$

$$= -\frac{1}{x^2} \sec\left(\frac{1}{x}\right) \tan\left(\frac{1}{x}\right)$$

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7. If $y = \cos^3 2x$, find y' Solution

$$y = \cos^3 2x$$

$$\text{Let } u = 2x \Rightarrow \frac{du}{dx} = 2$$

$$y = \cos^3 u = [\cos u]^3$$

$$\Rightarrow \frac{dy}{du} = 3[\cos u]^2 \cdot \sin u$$

$$\Rightarrow \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = -6 \cos^2 x \sin 2x$$

Notice $y = \cos^3 2x = [\cos 2x]^3$

$$y' = 3[\cos 2x]^2 \cdot (-2 \sin 2x)$$

$$= -6 \cos^2 2x \sin 2x$$

So, If $y = (3x^2 + x)^{10}$

$$\Rightarrow y' = 10(3x^2 + x)^9 (6x + 1)$$

$$= 10(3x^2 + x)^9 (6x + 1)$$

8. If $y = e^{2x} \ln x$, find y' Solution

$$y = e^{2x} \times \ln x \Rightarrow \text{product Rule.}$$

$$\text{let } u = e^{2x} \Rightarrow u' = 2e^{2x}$$

$$v = \ln x \Rightarrow v' = \frac{1}{x}$$

$$y' = u'v + v'u = 2e^{2x} \ln x + \frac{1}{x} e^{2x}$$

9. $f(x) = e^{\sqrt{1-x^2}}$, $f'(x) = ?$ (71)

Solution

$$y = f(x) = e^{\sqrt{1-x^2}}$$

$$\text{Let } u = \sqrt{1-x^2} = (1-x^2)^{\frac{1}{2}}$$

$$\frac{du}{dx} = \frac{1}{2}(1-x^2)^{-\frac{1}{2}} \cdot -2x$$

$$= \frac{-x}{(1-x^2)^{1/2}} = \frac{-x}{\sqrt{1-x^2}}$$

$$y = e^u = \frac{dy}{du} = e^u$$

$$\Rightarrow \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = \frac{-x}{\sqrt{1-x^2}} e^{\sqrt{1-x^2}}$$

10. $f(x) = \ln \left(\frac{x^2-x}{x^2-2x-8} \right)$, find $f'(x)$.

Solution

$$y = \ln \left(\frac{x^2-x}{x^2-2x-8} \right)$$

$$\text{let } u = \frac{x^2-2x}{x^2-2x-8} \Rightarrow \frac{du}{dx} = \frac{(2x-2)(x^2-2x-8) - (x^2-2x)(2x-2)}{(x^2-2x-8)^2}$$

$$y = \ln u \Rightarrow y' = \frac{1}{u} = \frac{x^2-2x-8}{x^2-2x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$= \frac{x^2-2x-8}{x^2-2x} \left[\frac{(2x-2)(x^2-2x-8) - (x^2-2x)(2x-2)}{(x^2-2x-8)^2} \right]$$

$$= \frac{(2x-2)[(x^2-2x-8) - (x^2-2x)]}{(x^2-2x)(x^2-2x-8)}$$

Exercise

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Find the derivative of the functions given

$$(a) f(x) = \cos(5x^3) \quad -15x^2 \sin(5x^3)$$

$$(b) f(x) = \sin^3 4x \quad 12 \sin^2 4x \cos 4x$$

$$(c) f(x) = (x^5 - 4x + 8)^7 \\ 7(x^5 - 4x + 8)^6 (5x^4 - 4)$$

$$(d) f(x) = \tan(\sec^2 x) \\ 2 \sec^2(\sec^2 x) \sec^2 x \tan x$$

$$(e) (3x+2) \cos 3x$$

$$(f) \sin(5x+2)$$

$$(g) f(x) = \left(\frac{3x^2 - 5}{2x^2 + 7} \right)^2$$

$$\frac{124x(3x^2 - 5)}{(2x^2 + 7)^3}$$

$$(h) f(x) = \frac{1 - \cos x}{1 + \cos x}$$